

WASP-52b

R-0459

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_170928 · created 2026-07-19T20:06:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458024.6466 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.206 +/- 0.012
Transit depth (Rp/Rs)^2	4.26 +/- 0.51 [%]
Orbital Inclination (inc)	85.71 +/- 0.71
Airmass coefficient 1 (a1)	0.9348 +/- 0.0082
Airmass coefficient 2 (a2)	0.059 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	#1 - [417, 228]
Optimal Aperture	4.1
Optimal Annulus	11.63
Transit Duration (day)	0.0658 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.206 ± 0.012 vs 0.1646 (deviation 3.45σ)
Duration fit vs published	0.0658 d vs 0.0758 d (deviation 4.01σ)
Mid-time O–C	-8.8 min
Depth SNR	17.2
Residual scatter	0.87 %

Light curve

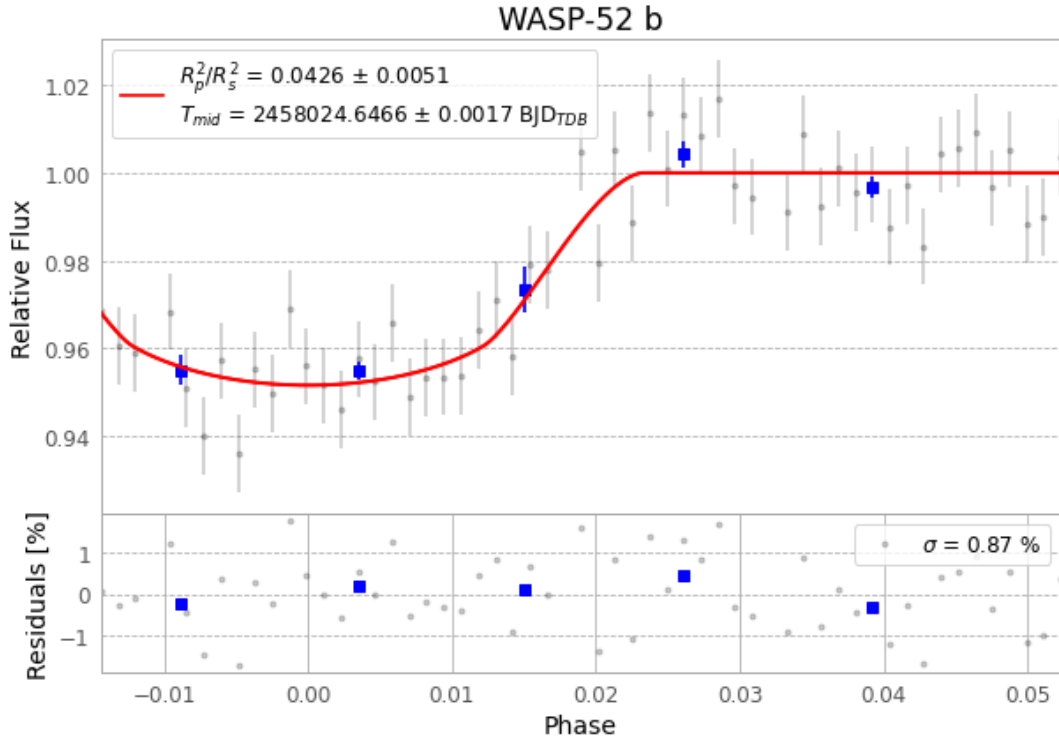


Figure 1 — FinalLightCurve_WASP-52 b_2017-09-27.png (SHA-256 e3d11b084c54a64a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 146], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e78ab1d3c9fc0a91...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

57 FITS frames (37 MB), WASP-52170928024512.FITS → WASP-52170928053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-170928.log (572aed9e20ba...), FinalLightCurve_WASP-52 b_2017-09-27.png (e3d11b084c54...), FinalLightCurve_WASP-52 b_2017-09-27.csv (f2ce5d8dcfb2...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 20:06Z.